

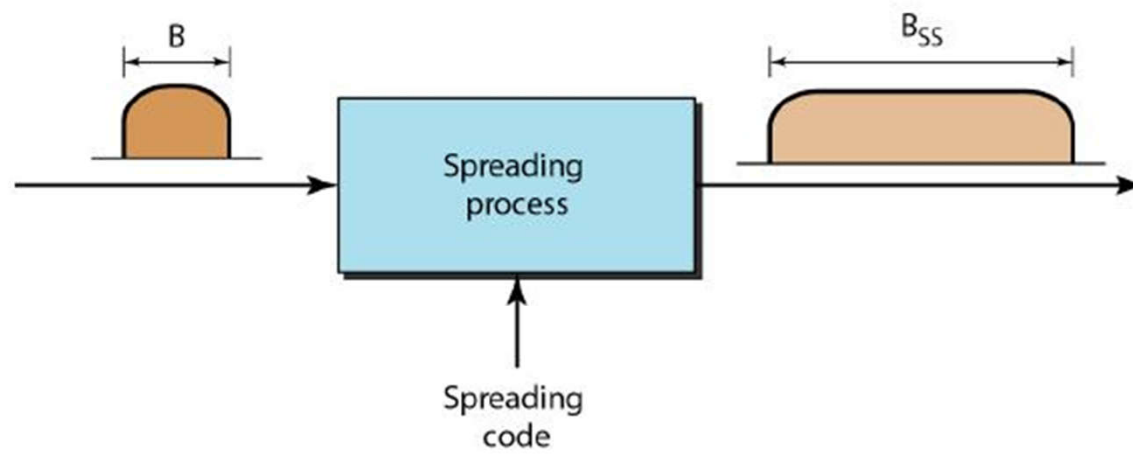
SPREAD SPECTRUM

Multiplexing combines signals from several sources to achieve bandwidth efficiency; the available bandwidth of a link is divided between the sources. In spread spectrum (88), we also combine signals from different sources to fit into a larger bandwidth, but our goals are somewhat different. Spread spectrum is designed to be used in wireless applications (LANs and WANs). In these types of applications, we have some concerns that outweigh bandwidth efficiency. In wireless applications, all stations use air (or a vacuum) as the medium for communication. Stations must be able to share this medium without interception by an eavesdropper and without being subject to jamming from a malicious intruder (in military operations, for example).

To achieve these goals, spread spectrum techniques add redundancy; they spread the original spectrum needed for each station. If the required bandwidth for each station is B , spread spectrum expands it to B_{ss} such that $B_{ss} \gg B$. The expanded bandwidth allows the source to wrap its message in a protective envelope for a more secure transmission. An analogy is the sending of a delicate, expensive gift. We can insert the gift in a special box to prevent it from being damaged during transportation, and we can use a superior delivery service to guarantee the safety of the package.

Figure 6.27 shows the idea of spread spectrum. Spread spectrum achieves its goals through two principles:

- A collective class of signaling techniques are employed before transmitting a signal to provide a secure communication, known as the **Spread Spectrum Modulation**. **The main advantage of spread spectrum communication technique is to prevent “interference” whether it is intentional or unintentional.**
- The signals modulated with these techniques are hard to interfere and cannot be jammed. An intruder with no official access is never allowed to crack them. Hence, these techniques are used for military purposes. These spread spectrum signals transmit at low power density and has a wide spread of signals.
- Pseudo-Noise (PN) Sequence
- A coded sequence of **1s and 0s** with certain auto-correlation properties, called as **Pseudo-Noise coding sequence** is used in spread spectrum techniques. It is a maximum-length sequence, which is a type of cyclic code.



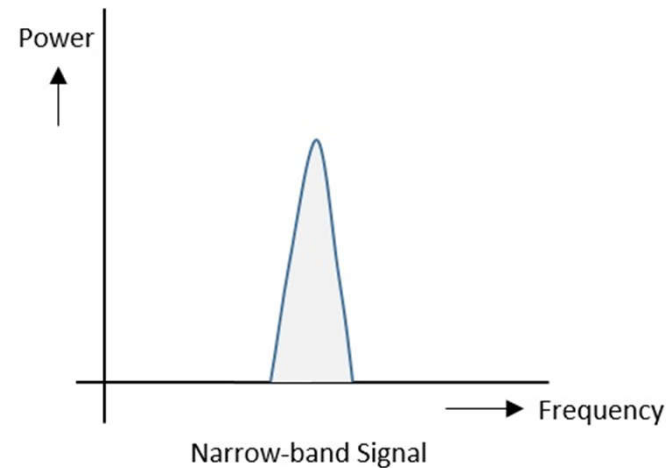
The spreading code is a series of numbers that look random, but are actually a pattern

Narrow-band and Spread-spectrum Signals

Both the Narrow band and Spread spectrum signals can be understood easily by observing their frequency spectrum as shown in the following figures.

Narrow-band Signals (compare with narrow lane)

The Narrow-band signals have the signal strength concentrated as shown in the following frequency spectrum figure.



Following are some of its features –

Band of signals occupy a narrow range of frequencies.

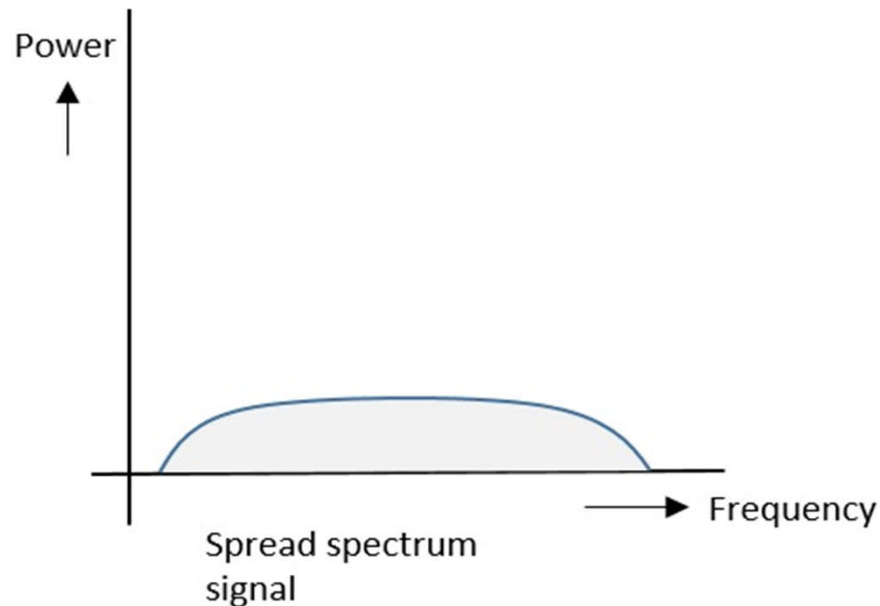
Power density (the amount of **power** (time rate of **energy** transfer) per unit volume) is high.

Spread of energy is low and concentrated.

Though the features are good, these signals are prone to interference.

Spread Spectrum Signals

The spread spectrum signals have the signal strength distributed as shown in the following frequency spectrum figure.



Following are some of its features –

Band of signals occupy a wide range of frequencies.

Power density is very low.

Energy is wide spread.

With these features, the spread spectrum signals are highly resistant to interference or jamming. Since multiple users can share the same spread spectrum bandwidth without interfering with one another, these can be called as **multiple access techniques**.

Spread spectrum multiple access techniques uses signals which have a transmission bandwidth of a magnitude greater than the minimum required RF bandwidth.

These are of two types.

1. Frequency Hopped Spread Spectrum FHSS

2. Direct Sequence Spread Spectrum DSSS

Frequency-hopping spread spectrum (FHSS) is a method of transmitting radio signals by rapidly changing the carrier **frequency** among many distinct **frequencies** occupying a large spectral band. (a wireless technology that spreads a signal over rapidly changing frequencies.) **FHSS** is used to avoid interference (the addition of unwanted signals to a useful signal.)(**Interference** occurs when unwanted radio frequency signals disrupt the use of your television, radio), to prevent eavesdropping, and to enable code-division multiple access (CDMA) communications.

Frequency Hopped Spread Spectrum FHSS

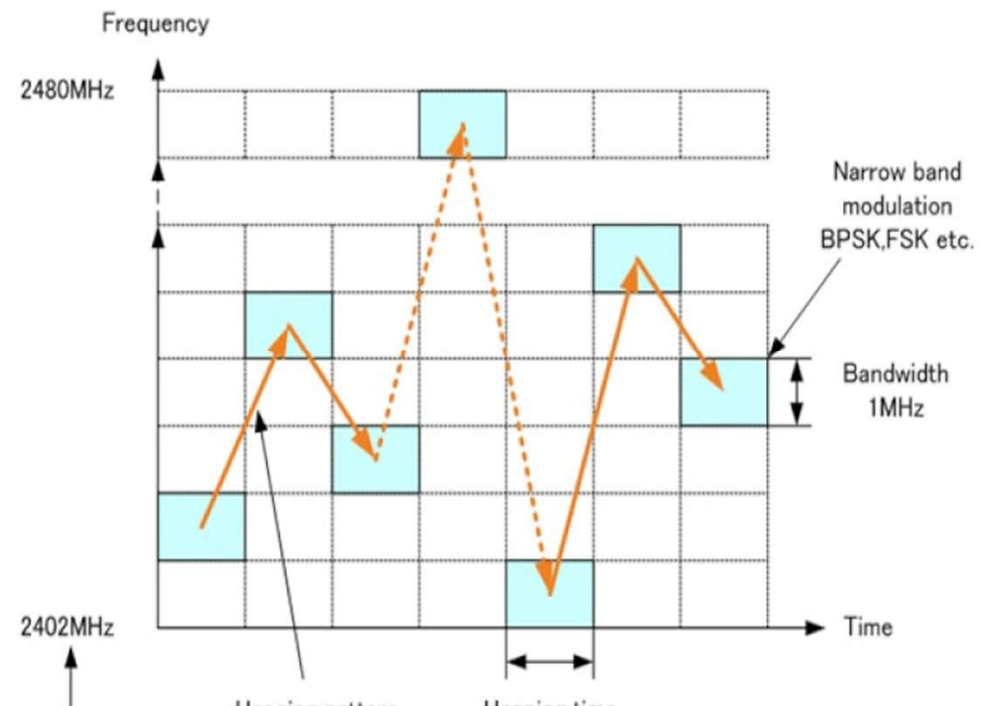
This is frequency hopping technique, where the users are made to change the frequencies of usage, from one to another in a specified time interval, hence called as **frequency hopping**. For example, a frequency was allotted to sender 1 for a particular period of time. Now, after a while, sender 1 hops to the other frequency and sender 2 uses the first frequency, which was previously used by sender 1. This is called as **frequency reuse**.

The frequencies of the data are hopped from one to another in order to provide a secure transmission. The amount of time spent on each frequency hop is called as **Dwell time(hop time)**.

Sender don't use single frequency to transmit data. Multiple frequency is used for transmission.

Sender send data using frequency f_1 for 625 micro second and then change frequency.

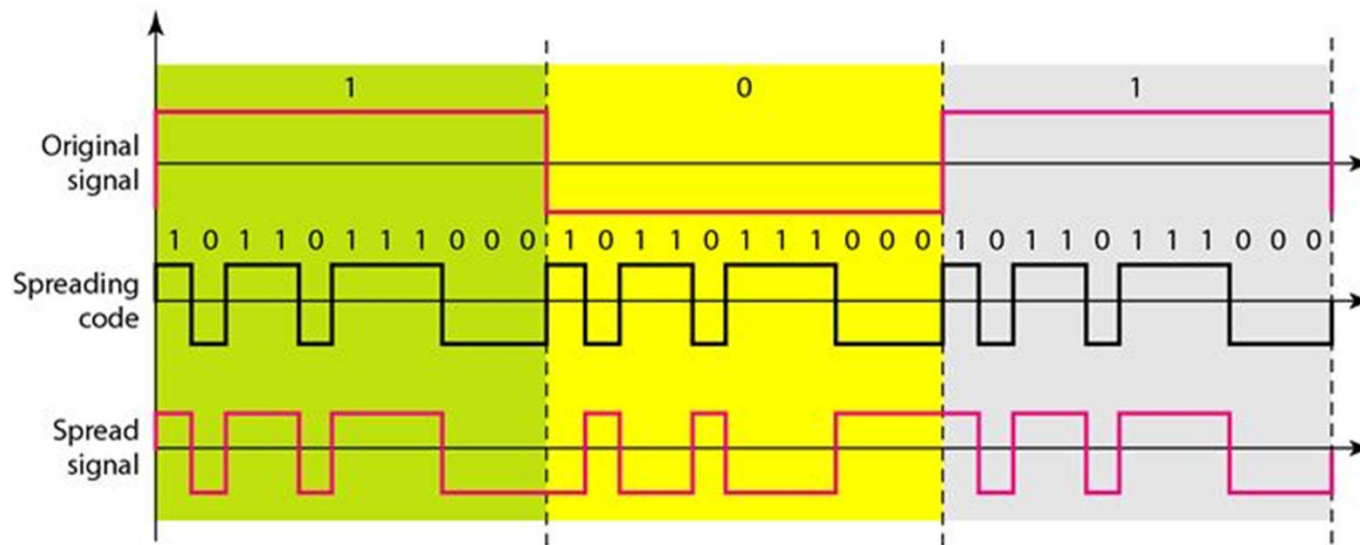
Different-2 sender use different-2 frequency patter.



Direct Sequence Spread Spectrum DSSS

Whenever a user wants to send data using this DSSS technique, each and every bit of the user data is multiplied by a secret code, called as **chipping code(spreading code)**. This chipping code is nothing but the spreading code which is multiplied with the original message and transmitted. The receiver uses the same code to retrieve the original message.

1 is represented by +1 and 0 is represented by -1. $1 \times 1 = 1$ and $1 \times -1 = -1$.



FHSS	DSSS / CDMA
Multiple frequencies are used	Single frequency is used
Hard to find the user's frequency at any instant of time	User frequency, once allotted is always the same
Frequency reuse is allowed	Frequency reuse is not allowed
Sender need not wait	Sender has to wait if the spectrum is busy
Power strength of the signal is high	Power strength of the signal is low
Stronger and penetrates through the obstacles	It is weaker compared to FHSS
It is never affected by interference	It can be affected by interference
It is cheaper	It is expensive
This is the commonly used technique	This technique is not frequently used